

$$\varphi(r_2) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_2} - \frac{1}{\infty} \right)$$

$$\varphi(r_2) = \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{r_2}$$

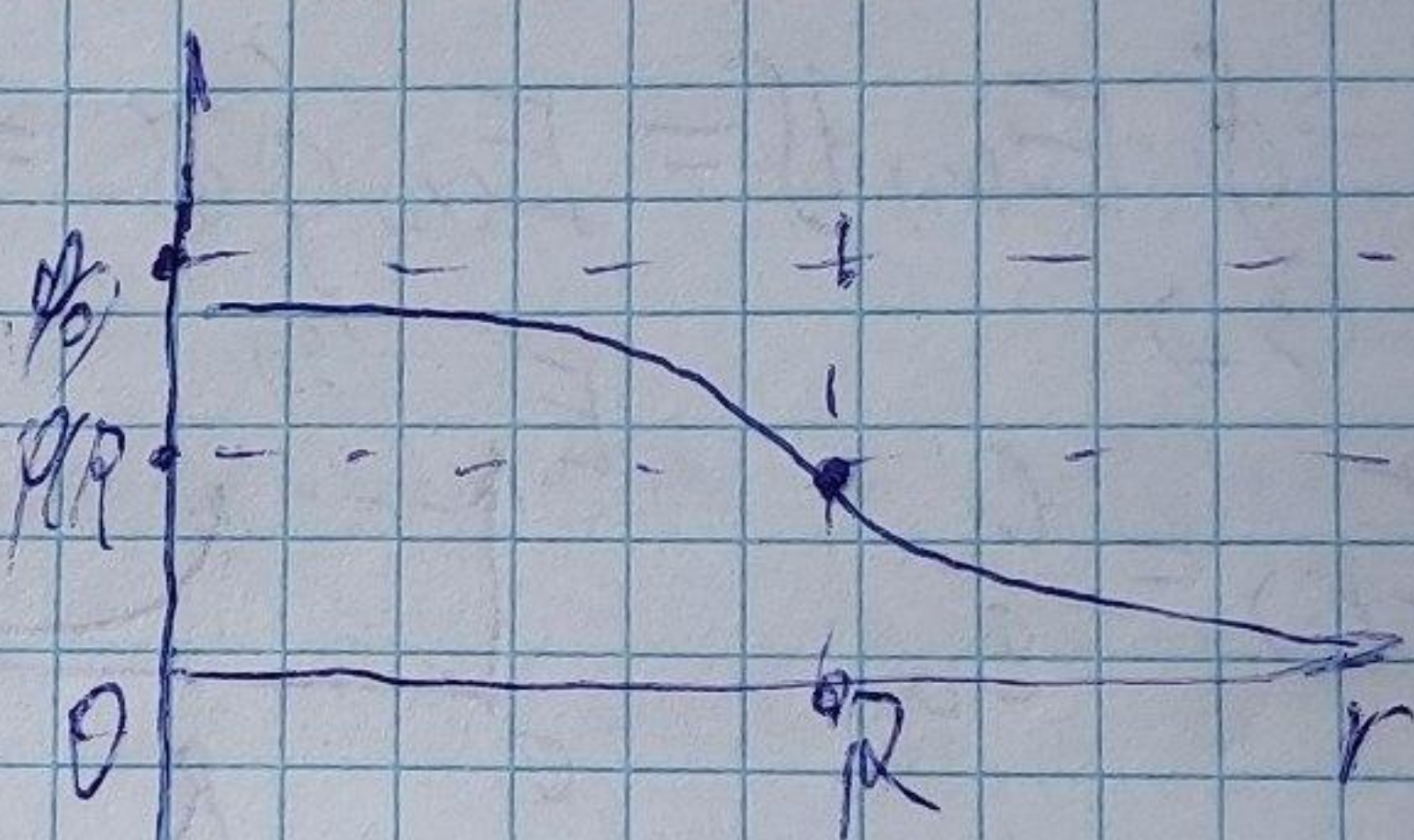
$$\int_{\varphi(R)}^{\varphi(r)} d\varphi = - \int_R^r \frac{Qr}{4\pi R^3 \epsilon_0} dr;$$

$$\varphi(r_1) - \varphi(R) = - \frac{Q}{4\pi R^3 \epsilon_0} \left(\frac{r^2}{2} \right) \Big|_R^{r_1}$$

$$\varphi(r_1) - \varphi(R) = - \frac{Q}{4\pi R^3 \epsilon_0} \left(\frac{r_1^2 - R^2}{2} \right); \quad \varphi(R) = \frac{Q}{4\pi\epsilon_0 R}$$

$$\varphi(r_1) = \frac{Q(3R^2 - r_1^2)}{8\pi R^3 \epsilon_0}$$

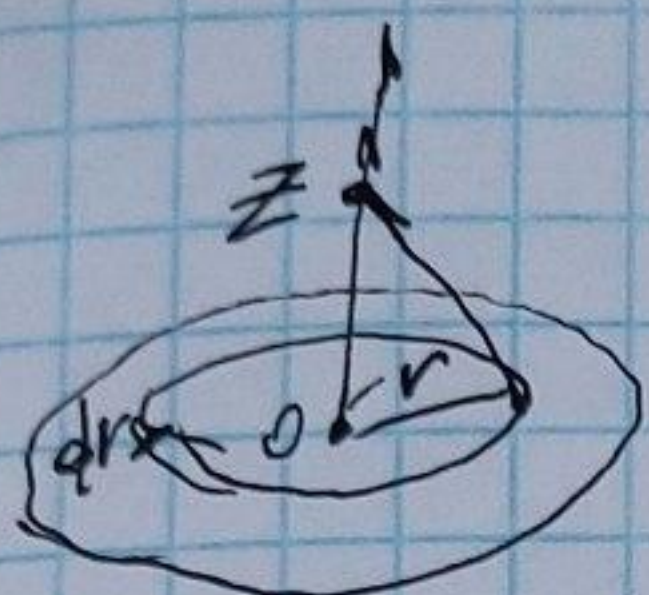
$$\varphi(0) = \frac{3Q}{8\pi R \epsilon_0}$$



$\vec{j} = \rho \vec{u}$; $\vec{j} = \sigma \vec{E}$ - закон Ома в дифференциальной форме
 $\vec{E} = -\vec{\nabla} \varphi$

$$\vec{j} = \sigma \vec{E} = e n \vec{u}$$

11.3.



$$a) d\varphi = \frac{k dq}{\sqrt{r^2 + z^2}}; \quad \sigma = \frac{d\varphi}{dS}; \quad dq = \sigma dS$$

$$\sigma = \frac{Q}{\pi R^2}; \quad dS = 2\pi r dr;$$

$$d\varphi = \frac{k \sigma dS}{\sqrt{r^2 + z^2}} = \frac{k Q \cdot 2\pi r dr}{\pi R^2 \sqrt{r^2 + z^2}};$$

$$\int_0^R d\varphi = \int_0^R \frac{k Q \cdot 2r dr}{R^2 \sqrt{r^2 + z^2}};$$

$$\varphi = \frac{2kQ}{R^2} \int \frac{r dr}{\sqrt{r^2 + z^2}} = \left| \begin{array}{l} u = r^2 + z^2 \\ du = 2r dr \end{array} \right. = \frac{2kQ}{R^2} \int_{z^2}^{R^2 + z^2} \frac{u^{-\frac{1}{2}} du}{2} =$$

$$= \frac{kQ}{R^2} \cdot u^{\frac{1}{2}} \cdot 2 \cdot \int_{z^2}^{R^2 + z^2} = \frac{2kQ}{R^2} (\sqrt{R^2 + z^2} - \sqrt{z^2})$$

$$b) \vec{E} = -\vec{\nabla} \varphi = -\bar{k} \frac{\partial \varphi(z)}{\partial z};$$

$$\vec{E} = -\bar{k} \frac{\partial \varphi}{\partial z} = -\bar{k} \left(\frac{2kQ}{R^2} (\sqrt{R^2 + z^2} - \sqrt{z^2}) \right) =$$

$$= -\bar{k} \cdot \frac{2kQ}{R^2} \left(\frac{z}{\sqrt{R^2 + z^2}} - 1 \right)$$